Deliverable #1: Software Requirement Specification (SRS)

SE 3A04: Software Design II – Large System Design

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1 Introduction

1.1 Purpose

The purpose of this document is to specify the functional and non-functional requirements of SCEMAS. This SRS will then serve as the guide for design and implementation for developers as well as other stakeholders involved. This will allow for a central truth when validating system behaviour and scope.

1.2 Scope

SCEMAS is a software system that supports environmental monitoring by accepting telemetry from distributed sensors, validating incoming messages, and storing accepted measurements for operational use and historical review. The system produces zone-based summaries of environmental conditions, such as short window averages and hourly peak values, and evaluates these results against administrator defined alert rules. When an alert condition is met, SCEMAS records the alert, maintains its status history, and makes the alert available to relevant users and interfaces.

The system supports multiple stakeholders through controlled access. Authorized operational users use SCEMAS to view current conditions, active alerts, and historical trends, and administrators use the system to manage alert rules and review audit relevant activity. SCEMAS also provides access to non-sensitive, aggregated environmental information for public and third party consumption through a read-only interface, with protections in place to maintain availability under heavy usage.

SCEMAS does not include physical sensor hardware, sensor installation, calibration, or maintenance. SCEMAS does not collect, store, or process personally identifiable information. User authentication is performed by an external authentication provider, and SCEMAS relies on that provider to verify user identity before granting access to protected functions.

1.3 Definitions, Acronyms, and Abbreviations

3PAS (Third-Party Authentication Service): An external identity provider used to authenticate users before SCEMAS grants access to protected functions.

Internet of Things (IoT): A network of physical objects ("things") embedded with sensors, software, and connectivity that enables them to collect and exchange data with other devices and systems.

Machine-to-Machine (M2M): Automated, direct exchange of data between devices, sensors, or machines without requiring human intervention.

Message Queuing Telemetry Transport (MQTT): A lightweight messaging protocol commonly used for Internet of Things and machine-to-machine communication.

Personally Identifiable Information (PII): Information that can identify an individual, including but not limited to name, address, phone number, or other identifying attributes.

Platform User: A user who accesses SCEMAS to view environmental information or alerts through the user-facing interfaces. Platform Users include Public Users, Third-Party Developers, and City Operators.

Role-Based Access Control (RBAC): An access control approach where permissions are assigned based on user roles, such as City Operator and System Administrator.

Smart City Environmental Monitoring & Alert System (SCEMAS): The software platform that ingests environmental telemetry, produces zone-based aggregates, and generates alerts for operators and the public.

SCEMAS Team: Authorized operational stakeholders responsible for monitoring, response coordination, and administrative oversight. This includes the System Administrator, City Operator, and Incident Response Team.

Transport Layer Security (TLS): A cryptographic protocol that provides secure communication over a computer network by encrypting data in transit.

User: Any person or system actor who interacts with SCEMAS. Users include City Operators, System Administrators, Incident Response Team members, Public Users, and Third-Party Developers.

1.4 References

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1.5 Overview

This document is organized into different sections describing SCEMAS. Section 1 introduces the system and outlines the purpose and scope of this SRS. Section 2 describes functions, user characteristics, constraints and assumptions. Section 3 presents use cases and actors a part of SCEMAS. Section 5 dives deeper into the system's non-functional requirements, while Section 6 describes new and innovative features in SCEMAS.

2 Overall Product Description

2.1 Product Perspective

SCEMAS is a cloud-native environmental monitoring and alerting platform intended to support smart city operations through real-time collection and analysis of environmental sensor telemetry. Similar products include commercial and research urban monitoring platforms that track air quality, noise, temperature, and weather conditions. SCEMAS brings these capabilities into a single system that supports day to day municipal decision making while also providing safe, aggregated information for public awareness. Rather than relying on sparse measurements or manual reporting, the system is designed around continuous streams of data from a distributed sensor network. Incoming measurements are checked for validity, stored for analysis, and transformed into operational insights that are easier to interpret than raw readings. SCEMAS also aggregates time-series data by predefined city zones so that metrics such as short-window averages and hourly peak values can be reported consistently and compared over time.

SCEMAS operates within a broader ecosystem of external components and consumers. On the input side, it receives sensor telemetry using the Message Queuing Telemetry Transport protocol, commonly referred to as MQTT, which is widely used for Internet of Things (IoT) style data collection [4]. Within the platform, validated telemetry persisted in storage suitable for time-series workloads, and the system uses zone-based grouping to generate meaningful summaries without exposing sensitive detail. On the output side, SCEMAS provides two primary interfaces. The first is a secure dashboard used by authorized users such as city operators and system administrators to view current conditions, trends, system health, and active alerts. The second is a public, read-only application programming interface that provides non-sensitive aggregated metrics for third-party applications, such as a digital signage display. SCEMAS also supports integration with subscribed external systems through an alert

notification endpoint so that downstream tools can respond automatically when alert conditions are met.

Access to protected functionality is supported through an external identity provider called the Third-Party Authentication Service, referred to as 3PAS, which authenticates users and enables role-based access control. Communications are protected using Transport Layer Security, referred to as TLS [6], including secure configurations for telemetry transport where appropriate. The platform follows privacy-by-design principles by avoiding the collection or storage of personally identifiable information within SCEMAS and prioritizing zone-level reporting that is suitable for public release [4].

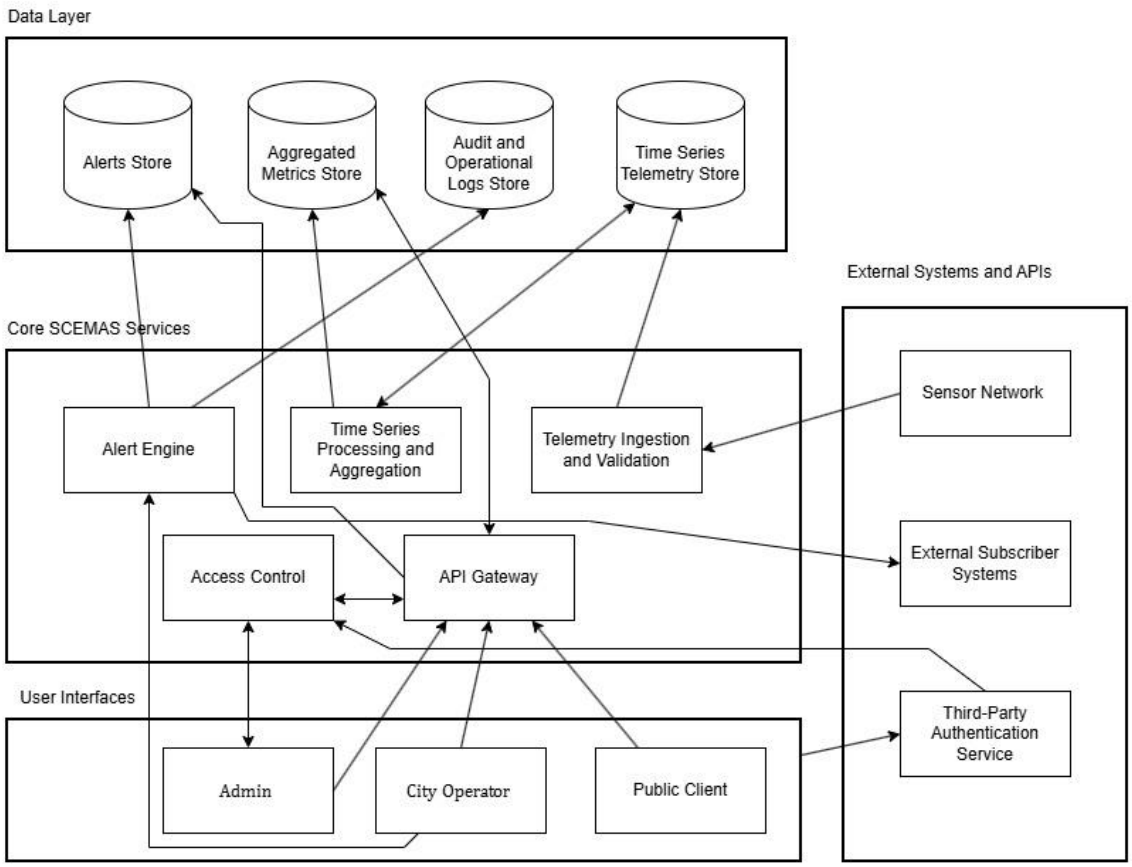

Figure 1. System Diagram

2.2 Product Functions

SCEMAS provides a set of major system functions that support continuous environmental monitoring, zone-based aggregation of time-series data, real-time alerting, and controlled access for different stakeholders. The functions below describe what the system does at a high level without stating detailed "shall" requirements.

Modules	Functions

Telemetry Intake and Validation	**Ingest Telemetry:** Receives environmental sensor telemetry over Message Queuing Telemetry Transport (MQTT). **Validate Telemetry Messages:** Confirms required fields, message structure, and plausible value ranges before accepting data.
Environmental Data Storage and Aggregation	**Store Validated Telemetry:** Persists accepted sensor readings in time-series storage for monitoring and historical review. **Compute Zone-Based Aggregates:** Groups readings by predefined zones and produces summaries such as short-window averages and hourly peak values. **Provide Historical Data Access:** Supports retrieval of past telemetry and aggregates to enable historical views by zone.
Alerting and Notification	**Manage Alert Rules:** Allows system administrators to define and update threshold based and trend based alert conditions. **Evaluate Conditions and Trigger Alerts:** Checks incoming readings and relevant aggregates to detect rule violations and generate alerts. **Track Alert Status and History:** Maintains alert lifecycle states such as active, acknowledged, and resolved, with history for review. **Notify Subscribed Parties:** Sends alert notifications to subscribed users and external subscriber systems through a notification endpoint.
Secure Access and Role Management	**Authenticate Users via Third-Party Authentication Service:** Uses the Third-Party Authentication Service (3PAS) to authenticate users before access is granted. **Authorize Access by Role:** Enforces role-based access control so users can access only permitted data and actions.
City Operator Dashboard	**Access Secured Dashboard Views:** Provides the city operator with access to dashboard views appropriate to their role after successful authentication through 3PAS and authorization through role-based access control. **Visualize Real-Time Conditions:** Displays current zone-level environmental conditions using maps, charts, and current-value indicators to support monitoring and decision-making. **Support Alert Operations:** Enables city operators to view alert details, acknowledge alerts, resolve alerts where permitted, and review alert history. **Display System Health:** Provides a view of system status and key operational indicators to support reliability monitoring.
Admin Dashboard	**Access Administrative Views:** Provides the system administrator with access to administrative functions after successful authentication through 3PAS and authorization through role-based access control. **Manage Alert Rules:** Enables administrators to create, update, and activate

	or deactivate alert rules, including threshold conditions, trend conditions, and zone scope. **Review Alerts with Rule Context:** Allows administrators to view full alert details including the rule conditions that were violated and the relevant threshold context. **Review Audit and Configuration Activity:** Provides access to records of authentication events and configuration changes relevant to system oversight.
Public and Third-Party Data Access	**Expose Read-Only Public Data Interface:** Provides a read-only application programming interface for non-sensitive aggregated environmental information by zone. **Apply Request Limiting:** Limits request rates on public endpoints to reduce abuse and maintain availability. **Provide API Documentation:** Supplies documentation sufficient for third-party developers to request and interpret environmental data. **Provide Demonstration Public Client:** Provides a simple client, such as a digital signage display, that consumes the public interface.
Security, Privacy, and Auditability	**Protect Communications:** Uses Transport Layer Security (TLS) to protect data in transit, including secure telemetry transport where appropriate. **Preserve Privacy by Design:** Avoids collecting or storing personally identifiable information and limits public outputs to zone-level aggregates. **Record Audit and Operational Events:** Logs significant events such as authentication activity, configuration changes, and alert lifecycle events for accountability.

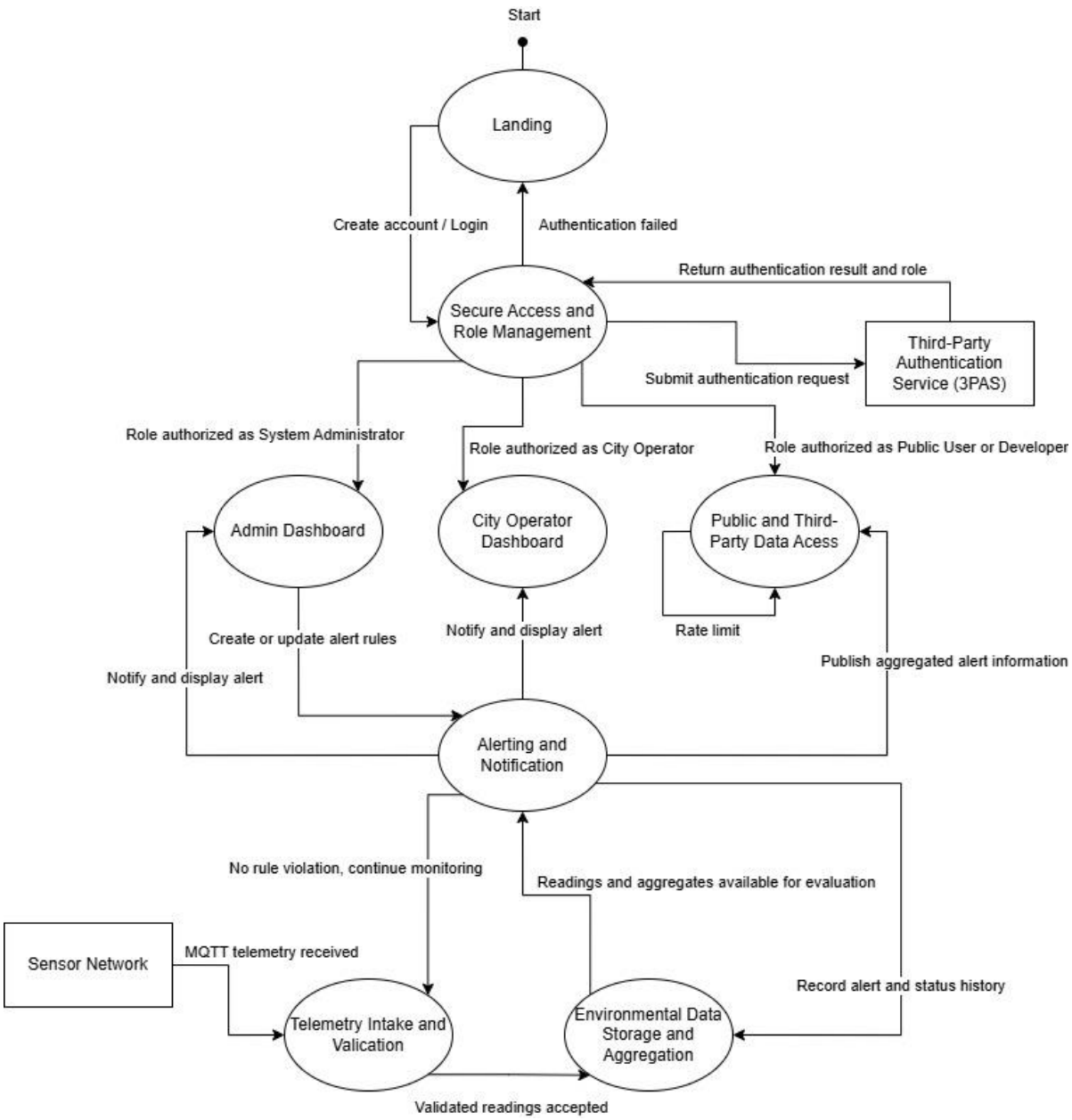

Figure 2. State Diagram

2.3 User Characteristics

SCEMAS supports multiple user types. Each user type differs in expected experience, technical expertise, and responsibilities. The system's interfaces and access privileges are tailored to these characteristics through role-based access control.

City Operator:

- Education level: Municipal staff training and basic computer literacy.
- Experience: Familiar with operational dashboards and responding to alerts during routine monitoring and incident conditions.
- Technical expertise: Comfortable using a web dashboard, interpreting maps and charts, and following standard authentication steps. Not expected to have software development knowledge.

System Administrator:

- Education level: Technical or operational administration background appropriate for managing system configuration.
- Experience: Experienced with system administration tasks such as configuring rules, reviewing logs, and overseeing access permissions.
- Technical expertise: Comfortable with administrative dashboards, role-based access concepts, and interpreting audit or operational logs. Not expected to modify source code as part of normal use.

Incident Response Team:

- Education level: Emergency response or municipal operations background.
- Experience: Engages primarily during significant events; experienced in reviewing incident summaries and issuing guidance to affected regions.
- Technical expertise: Able to interpret alert information and recommended actions using a web interface. Does not require deep technical knowledge of data pipelines or system internals.

Public User:

- Education level: General public; no specialized education assumed beyond basic literacy.
- Experience: May be a first-time user; may access the system infrequently and primarily during major events.
- Technical expertise: Assumed to have basic ability to navigate a web or mobile interface. Public-facing information should be easy to understand and presented using plain language.

Third-Party Developer:

- Education level: Software development education or equivalent professional experience.
- Experience: Familiar with integrating external services and consuming application programming interfaces.
- Technical expertise: Comfortable working with REST APIs, authentication workflows, and structured data formats such as JSON. Not expected to have domain expertise in environmental science, so documentation must support correct interpretation of zone-level aggregated data.

2.4 Constraints

The development of SCEMAS is subject to constraints that limit implementation choices and influence the final scope of the system.

1. Privacy by Design

- SCEMAS must be designed with privacy in mind from the start [4]. It must not collect, store, or expose any personally identifiable information, and anything shown to the public must remain non-sensitive and aggregated by predefined city zones. This limits how detailed the data can be in storage, dashboards, logs, alerts, and public interfaces, since the system must avoid revealing identifying information in any form.

2. Project Time:
   - The time available to build SCEMAS limits how large and complex the system can be within the project period. As a result, the team must focus on delivering the core functions first, such as telemetry ingestion, aggregation, alerting, secure access, and public data access, and leave lower-priority enhancements for future iterations if needed.

2.5 Assumptions and Dependencies

Assumptions:

1. Sensor devices publish telemetry using the expected message structure, required fields, and measurement units so that SCEMAS can validate and aggregate readings consistently.
2. The city is divided into predefined zones, and those zones are available to SCEMAS for grouping readings and producing public, non-sensitive summaries.
3. Users have reliable network access when using the operator dashboard, administrator dashboard, and public interfaces.
4. The alert rules defined by administrators are meaningful and based on agreed environmental thresholds or trends, and they are updated when city policy or environmental standards change.
5. SCEMAS can meet monitoring and alerting objectives without storing personally identifiable information, since privacy-by-design is a core constraint of the system.
6. External subscriber systems that receive alert notifications are capable of receiving and processing notifications in the expected format.

Dependencies:

1. SCEMAS depends on the Third-Party Authentication Service (3PAS) for user authentication and the identity verification needed to enforce role-based access control.
2. SCEMAS depends on the availability of a secure transport mechanism such as Transport Layer Security (TLS) for protecting communications, including secure configurations for telemetry transport where applicable [6].
3. SCEMAS depends on the messaging infrastructure used for telemetry transport to be available and stable so that sensor data can be received without frequent interruption [5].

Other assumptions that, if they fail to hold, could require a change to the requirements:

1. If sensor payload formats, field definitions, or units are inconsistent across devices or change over time, SCEMAS would require updated validation rules and schema handling.
2. If zone boundaries are not stable, or if reporting must be performed at a finer granularity than zones, the aggregation approach and privacy constraints would need to be revisited.
3. If 3PAS cannot support the required roles or experiences extended downtime, the login and access workflows described in this SRS would need to change.
4. If stakeholders require user-specific data within SCEMAS (beyond non-identifying preferences), privacy and data handling requirements would need to be expanded and redefined.

2.6 Apportioning of Requirements

The initial release of SCEMAS will prioritize the core monitoring workflow, including telemetry ingestion and validation, time-series storage and zone-based aggregation, rule-based alerting, secure access through 3PAS with role-based access control, audit logging, and the public read-only API with rate limiting. The following items may be deferred to later versions if scope or complexity requires it.

1. Advanced trend and anomaly detection:

- The initial release will focus on clear threshold-based alerting. More complex trend-based or anomaly-based detection can be introduced in later versions.

2. Expanded notification delivery channels

- The initial release will rely on dashboard visibility and the alert notification endpoint for subscribed systems. Additional channels such as SMS and mobile push notifications may be added later.

3. Enhanced historical analytics and reporting

- The initial release will support basic historical viewing needed for dashboards and public summaries. More advanced reporting features such as exports, long-range comparisons, and automated summaries may be deferred.

3 Use Case Diagram

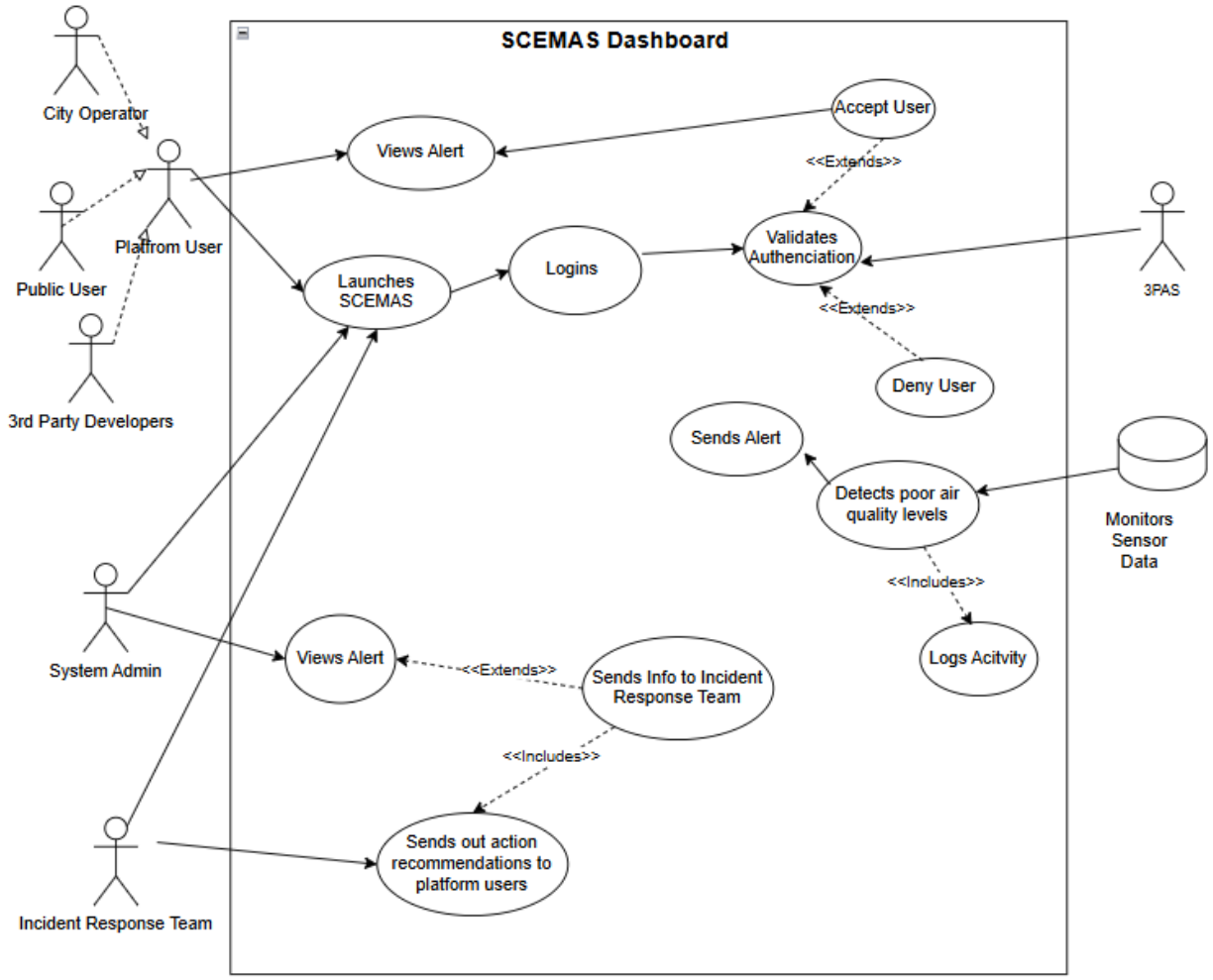

Figure 3. Use Case Diagram

4 Highlights of Functional Requirements

The business events we will consider:

- BE1: Pollution causing air quality to be extremely poor.
- BE2: Ongoing pollution alert has ended, and air quality has returned to normal.
- BE3: Scheduled Environmental Sampling Taken
- BE4: Create an Account
- BE5: User Logs into Dashboard
- BE6: Define/Create alert rules

The Viewpoints we will consider are:

- VP1: City Operators
- VP2: Public User
- VP3: Third-Party Developer
- VP4: System Administrator
- VP5: Incident Response Team

BE1. Pollution occurs causing air quality to be extremely poor.

Pre-Conditions:

- City Environment Threshold Exceeded
- System is in a functional state and was not in an alerting state before threshold was exceeded
- Public User, and third-party developer are subscribed
- Incidence report Department is on duty and logged in.

VP1. City Operator
Main Success Scenario

1. Air quality sensor readings cross environment thresholds.
2. System's Alert Engine sends out a notification regarding the alert.
3. City operator receives alert.
4. City operator opens SCEMAS application.
5. System launches Login/Account Creation Page.
6. City Operator clicks on Login Page and selects the "City Operator" button.
7. System redirects to 3PAS portal.
8. City Operator enters login credentials.
9. System receives authentication verification confirmation.
10. System presents the city operator with all details relating to the alert.
11. City operator can now review system health, alert details, and historic data trends.

Secondary Success Scenario

2.1.i Alert Engine notification fails to send to the city operator.
2.1.ii System re-sends the notification.

10.1.i Alert details fail to be presented.
10.1.ii Database is re-queried for relevant information.

VP2. Public User
Main Success Scenario

3.b Public user receives the alert.
4.b Public user opens the SCEMAS application.
6.b Public User clicks on Login Page and selects the "Public User" button.
8.b Public User enters login credentials.
10.b System presents the public user with aggregated and non-sensitive details relating to the alert.
11.b Public User can now view aggregated data to better understand the alert.

Secondary Success Scenario

2.1.i Alert Engine notification fails to send to the public user.
2.1.ii System re-sends the notification.

10.1.i Alert details fail to be presented.
10.1.ii Database is re-queried for relevant information.

VP3. Third-Party Developer

Main Success Scenario

1. Air quality sensor readings cross environment thresholds.
2. System's Alert Engine notifies the third-party developer through notifier that an alert was issued.
3. Developer receives the alert.
4. Developer launches their application.
5. Developer's application sends a request for data to the SCHEMA API endpoint.
6. System receives the request and asks 3PAS to verify the developer.
7. Developer's application provides required authentication credentials.
8. 3PAS successfully authenticates the credentials.
9. System sends data packets of aggregated, non-sensitive data relating to the alert.

Secondary Success Scenario

5.1.i HTTPS request fails to send.
5.1.ii Developer re-sends the request.

VP4. System Admin

3.b System Admin receives the alert.
4.b System Admin opens the SCEMAS application.
6.b System Admin clicks on Login Page and selects the "System Admin" button
8.b System Admin enters login credentials.
10.b System Admin can now view all details of the alert through the admin dashboard and view what thresholds were exceeded.
11.b System Admin refers necessary information to the Incidence Response Department

VP5. Incident Response Department

1. Incidence Response Department receives details of an alert from the system admin.
2. Incidence Response Department reviews the details.
3. Incidence Response Department plans zone-based course of action for individuals of every zone.
4. Incidence Response Department sends course of action to all subscribed users through alert engine.

Global Scenario:

*The term account holder will be used to refer to VP2, VP3
For simultaneous events #.a, b, d, and d will be used for viewpoints VP1, VP2, VP3, VP4

Main Success Scenario

1. Air quality sensor readings cross environment thresholds that were set up by the System Admin.
2. System updates the log with this event.
3. System's Alert Engine sends out a notification regarding the alert.
4. Account holders and SCEMAS team receives the alert.

5.a Public Users launch SCEMAS software.
5.b City Operators launch SCEMAS software.
5.d System Admins launch SCEMAS software.

6. System launches Login/Account Creation Page.

7.a Public Users clicks on Login Page and selects the "Public User" button.
7.b City Operators clicks on Login Page and selects the "City Operator" button.
7.d System Admins clicks on Login Page and selects the "System Admin" button.

8. System redirects to 3PAS portal.
9. Account holders enter their authentication details.
10. 3PAs successfully authenticates the account holder.

11.a System presents the city operator with all details relating to the alert.
11.b System presents the public user with aggregated and non-sensitive details relating to the alert.
11.d System Admin can view all details of the alert through the admin dashboard and view what thresholds were exceeded.

12. System Admin refers necessary information to the incident Response Department.
13. Incidence Response Department receives details of an alert from the system admin.
14. Incidence Response Department reviews the details.
15. Incidence Response Department plans zone-based course of action for individuals of every zone.
16. Incidence Response Department sends course of action to all subscribed users through alert engine.
17. Account holders receive zone-based course of actions.

Secondary Success Scenario

3.1.i Alert Engine notification fails to be sent.
3.1.ii System re-sends the notification.

11.1.i Alert details fail to be presented.
11.1.ii Database is re-queried for relevant information.

BE2. Ongoing pollution alert has ended, and air quality has returned to normal

Pre-Conditions:

- System is functional and was previous in an alert state.

- 3PAS contains data of all account holders and SCHEMA TEAM.
- City Operator, Public User, and third-party developer are subscribed.
- Incidence report Department is on duty and logged in.

VP1. City Operator
Main Success Scenario

1. Air quality sensor readings have transitioned to be under environmental threshold.
2. System's Alert Engine sends out a notification through the notifier regarding how the alert is no longer active.
3. City operator receives alert.
4. City operator opens SCEMAS application.
5. System launches Login/Account Creation Page.
6. City Operator clicks on Login Page and selects the "City Operator" button.
7. System redirects to 3PAS portal.
8. City Operator enters login credentials.
9. System receives authentication verification confirmation.
10. System presents the city operator with detailed overview of concluded alert and system health.

Secondary Success Scenario

2.1.i Alert Engine notification fails to send to the city operator.
2.1.ii System re-sends the notification.

10.1.i Alert details fail to be presented.
10.1.ii Database is re-queried for relevant information.

VP2. Public User
Main Success Scenario

3.b Public user receives the alert.
4.b Public user opens the SCEMAS application.
6.b Public User clicks on Login Page and selects the "Public User" button.
8.b Public User enters login credentials.
10.b System presents the public user with a detailed confirmation that the alert is no longer active and aggregated data relating to the alert.

Secondary Success Scenario

2.1.i Alert Engine notification fails to send to the public user.
2.1.ii System re-sends the notification.

10.1.i Alert details fail to be presented.
10.1.ii Database is re-queried for relevant information.

VP3. Third-Party Developer

Main Success Scenario

1. Air quality sensor readings have transitioned to be under environmental threshold
2. System's Alert Engine sends out a notification through the notifier regarding how the alert is no longer active.
3. Developer receives the alert.
4. Developer launches their application.
5. Developer's application sends a request for data to the SCHEMA API endpoint.
6. System receives the request and asks 3PAS to verify the developer.
7. Developer's application provides required authentication credentials.
8. 3PAS successfully authenticates the credentials.
9. System sends data packets containing the details of the concluded alert.

Secondary Success Scenario

5.1.i HTTPS request fails to send.

5.1.ii Developer re-sends the request.

VP4. System Admin

3.b System Admin receives the alert.

4.b System Admin opens the SCEMAS application.

6.b System Admin clicks on Login Page and selects the "System Admin" button.

8.b System Admin enters login credentials.

10.b System Admin can view all details of the concluded alert through the system admin dashboard and view detailed information regarding the health of the system.

11.b. System Admin refers necessary information to the incident Response Department.

VP5. Incident Response Department

1. Incidence Response Department receives details of the concluded alert from the system admin.
2. Incidence Response Department reviews the details.
3. Incidence Response Department creates a course of action for individuals who may have been affected by the pollution.
4. Incidence Response Department sends course of action to all subscribed users through the Alert Engine.

Global Scenario:

*The term account holder will be used to refer to VP2, VP3

For simultaneous events #.a, b, and c will be used for viewpoints VP1, VP2, VP3

Main Success Scenario

1. Air quality sensor readings fall under thresholds that were set up by the System Admin and were previously crossed.
2. System updates the log with the end of the ongoing alert.
3. System's Alert Engine sends out a notification through the notifier regarding how the alert is no longer active.
4. Account holders and SCEMAS team receives the alert.

5.a Public Users launch SCEMAS software.
5.b City Operators launch SCEMAS software
5.d System Admins launch SCEMAS software

6. System launches Login/Account Creation Page.

7.a Public Users clicks on Login Page and selects the "Public User" button.
7.b City Operators clicks on Login Page and selects the "City Operator" button.
7.d System Admins clicks on Login Page and selects the "System Admin" button.

8. System redirects to 3PAS portal.
9. Account holders enter their authentication details.
10. 3PAs successfully authenticates the account holder.

11.a System presents the city operator with detailed overview of concluded alert and system health.
11.b System presents the public user with a detailed confirmation that the alert is no longer active and aggregated data relating to the alert.
11.d System Admin can view all details of the concluded alert through the system admin dashboard and view detailed information regarding the health of the system.

12. System Admin refers necessary information to the incident Response Department.
13. Incidence Response Department reviews the details.
14. Incidence Response Department creates a course of action for individuals who may have been affected by the pollution.
15. Incidence Response Department sends course of action to all subscribed users through the Alert Engine.
16. Account holders receive zone-based course of actions.

Secondary Success Scenario

3.1.i Alert Engine notification fails to be sent
3.1.ii System re-sends the notification.

11.1.i Concluded alert details fail to be presented.
11.1.ii Database is re-queried for relevant information.

BE3. Scheduled Environmental Sample Taken

Pre-Conditions:

- System is functional and was previous in an alert state
- There exists a populated developer registry, city operator registry, and public user registry
- City Operator, Public User, and third-party developer are subscribed
- Incidence report Department is on duty and logged in.

VP1. City Operator

Main Success Scenario

1. City operator opens SCEMAS application.
2. System launches Login/Account Creation Page.
3. City Operator clicks on Login Page and selects the "City Operator" button.
4. System redirects to 3PAS portal.
5. City Operator enters login credentials.
6. System receives authentication verification confirmation.
7. System presents the city operator with the main dashboard displaying visualized environmental metrics, and system state details from the up-to-date system database.

Secondary Success Scenario

7.1.i Data fails to be fetched from database.
7.1.ii Database is re-queried for relevant information.

VP2. Public User

Main Success Scenario

1.b Public user opens the SCEMAS application.
3.b Public User clicks on Login Page and selects the "Public User" button.
5.b Public User enters login credentials.
7.b System presents a read only public dashboard displaying aggregated, not-sensitive environmental data from up-to-date system database.

Secondary Success Scenario

7.1.i Data fails to be fetched from database.
7.1.ii Database is re-queried for relevant information.

VP3. Third-Party Developer

Main Success Scenario

1. Developer launches their application.
2. Developer's application sends a request for data to the SCHEMA API endpoint.
3. System receives the request and asks 3PAS to verify the developer.
4. Developer's application provides required authentication credentials.
5. 3PAS successfully authenticates the credentials.
6. System sends data packets of aggregated, non-sensitive data.

Secondary Success Scenario

2.1.i HTTPS request fails to send.
2.1.ii Developer re-sends the request.

VP4. System Admin

N/A

VP5. Incident Response Department

N/A

Global Scenario:

*The term account holder will be used to refer to VP2, VP3
For simultaneous events #.a, b, and c will be used for viewpoints VP1, VP2, VP3

Main Success Scenario

1. Ample time has passed since previous data collection period so new sample is taken.
2. Current data readings are updated as well as aggregated data collections.

3.a Public Users launch SCEMAS software.
3.b City Operators launch SCEMAS software

4. System launches Login/Account Creation Page.

5.a Public Users clicks on Login Page and selects the "Public User" button.
5.b City Operators clicks on Login Page and selects the "City Operator" button.

6. System redirects to 3PAS portal.
7. Account holders enter their authentication details.
8. Account holders enter their authentication details.
9. 3PAs successfully authenticates the account holder.

11.a System presents the city operator with the main dashboard displaying visualized environmental metrics, and system state details from the up-to-date system database.
11.b System presents a read only public dashboard displaying aggregated, not-sensitive environmental data from up-to-date system database.

Secondary Success Scenario

11.1.i Data fetch from database fails.
11.1.ii Database is re-queried for relevant information.

BE4 : Create An Account
Pre-condition - No viewpoints have an existing account

VP1: City Operator #1
Main Success Scenario:

1. City Operator opens SCEMAS application to gain access.
2. System launches Login/Account Creation portal.
3. City Operator clicks on "create new account" and selects role as "City Operator".

4. The system re-directs the city operator to a 3PAS portal.

5. City Operator enters required information to register account.
6. The system receives confirmation that the authentication portal has verified and validated the information.
7. City Operators redirected back to the SCEMAS account creation page.
8. City Operator adds user preferences (communication/subscription preferences).
9. System saves preferences.
10. City Operators have access to SCEMAS Dashboard.

Secondary scenario:

4.1.i System fails to redirect to 3PAS portal.
4.1.ii System re-launches account creation/login page.

5.1.i City Operator adds incorrect credentials.
5.1.ii Prompted to retry.
5.1.iiii If authentication fails again, session timed out and the city operator does not have access to SCEMAS dashboard.

6.1.i The system received verification was unsuccessful.
6.1.ii Session timed out, system prompts to try again and re-directs to homepage.
6.1.iii City operator unable to create account.

7.1.i System unable to re-direct to SCEMAS account creation page.
7.1.ii System reloads current page.

8.1.i System unable to save user preferences.
8.1.ii System reloads page prompts user to try saving preferences again.

VP2: Public User #2

1b. Public User opens SCEMAS portal to gain access
3b. Public User clicks on "create new account" and selects role as "Public User".
5b. Public User enters required information to register account.
7b. Public User redirected back to the SCEMAS account creation page.
8b. Public User adds user preferences (communication/subscription preferences).

Secondary scenario:

4.1.i System fails to redirect to 3PAS portal
4.1.ii System re-launches account creation/login page

5.1.i Public User adds incorrect credentials.
5.1.ii Prompted to retry.
5.1.iiii If authentication fails again, session timed out and the public user does not have access to SCEMAS dashboard.

6.1.i The system received verification was unsuccessful.
6.1.ii Session timed out, system prompts to try again and re-directs to homepage.
6.1.iii Public User unable to create account.

7.1.i System unable to re-direct to SCEMAS account creation page.
7.1.ii System reloads current page.

8.1.i System unable to save user preferences
8.1.ii System reloads page prompts user to try saving preferences again.

VP3: Third Party Developer #3

1. Third Party Developer opens SCEMAS application to gain access
2. System launches Login/Account Creation portal
3. Third Party Developer clicks on "create new account" and selects role as "Third Party Developer"
4. The system re-directs the third-party developer to a 3PAS portal.
5. Third Party Developer enters required information to register account.
6. 3PAS provides developer with credentials (API KEY).

Secondary scenario:

4.1.i System fails to redirect to 3PAS application.
4.1.ii System re-launches account creation/login page.

5.1.i Third Party Developer adds incorrect credentials.
5.1.ii Prompted to retry.
5.1.iiii If authentication fails again, session timed out and the system admin does not have access to SCEMAS dashboard.

6.1.i The system received verification was unsuccessful.
6.1.ii Session timed out, system prompts to try again and re-directs to homepage.
6.1.iii Third Party Developer unable to create account.

VP4: System Administrator #4

1d. System Administrator open SCEMAS application to gain access.
3d. System Administrator clicks on "create new account" and selects role as "System Administrator".
5d. System Administrator enters required information to register account.
7d. System Administrator redirected back to the SCEMAS account creation page.
8d. System Administrator adds user preferences (communication/subscription preferences).

Secondary scenario:

4.1.i System fails to redirect to 3PAS portal.
4.1.ii System re-launches account creation/login page.

5.1.i System Administrator adds incorrect credentials.
5.1.ii Prompted to retry.
5.1.iiii If authentication fails again, session timed out and the system administrator does not have access to SCEMAS dashboard.

6.1.i The system received verification was unsuccessful.
6.1.ii Session timed out, system prompts to try again and re-directs to homepage.
6.1.iii System Administrator unable to create account.

7.1.i System unable to re-direct to SCEMAS account creation page.
7.1.ii System reloads current page.

8.1.i System unable to save user preferences.
8.1.ii System reloads page prompts user to try saving preferences again.

VP5 : Incident Response Team #5

1e. Incident Response Team open SCEMAS application to gain access.
3e. Incident Response Team clicks on "create new account" and selects role as "Incident Response Team".
5e. Incident Response Team enters required information to register account.
7e. Incident Response Team back to the SCEMAS account creation page.
8e. Incident Response Team adds user preferences. (communication/subscription preferences)

Secondary scenario:

4.1.i System fails to redirect to 3PAS portal.
4.1.ii System re-launches account creation/login page.

5.1.i Incident Response Team adds incorrect credentials.
5.1.ii Prompted to retry.
5.1.iiii If authentication fails again, session timed out and the Incident Response Team does not have access to SCEMAS dashboard.

6.1.i The system received verification was unsuccessful.
6.1.ii Session timed out, system prompts to try again and re-directs to homepage.
6.1.iii Incident Response Team unable to create account.

7.1.i System unable to re-direct to SCEMAS account creation page.
7.1.ii System reloads current page.

8.1.i System unable to save user preferences.

8.1.ii System reloads page prompts user to try saving preferences again.

Global Scenario:
Pre-condition: No viewpoints have an existing account
Main Success Scenario:

1. User opens SCEMAS application to gain access.
2. System launches Login/Account Creation portal.
3. User clicks on "create new account" and selects relevant role.
4. The system re-directs the user to a 3PAS authentication portal.
5. User enters required information to register account.
6. The system receives confirmation that the authentication portal has verified and validated the information.
7. User redirected back to the SCEMAS account creation page.
8. User adds user preferences (communication/subscription preferences).
9. System saves preferences.
10. User have access to SCEMAS Dashboard.

Secondary scenario:

4.1.i System fails to redirect to relevant role page.
4.1.ii System relaunches account creation/login page.

5.1.i User adds incorrect credentials.
5.1.ii Prompted to retry.
5.1.iii If authentication fails again, session timed out and the user does not have access to SCEMAS dashboard.

6.1.i The system received verification was unsuccessful.
6.1.ii Session timed out, system prompts to try again and re-directs user to homepage.
6.1.iii User unable to create account.

7.1.i. System unable to re-direct to SCEMAS account creation page
7.1.ii System re-loads current page.

8.1.i System unable to save user preferences.
8.1.ii System re-loads page prompts user to try saving preferences again.

BE5: User Logins into Dashboard

Pre-condition - Viewpoints have an existing account with SCEMAS
VP1 : City Operator #1

1. City Operator opens SCEMAS application.
2. System launches Login/Account Creation Page.
3. City Operator clicks on Login Page, selects "City Operator" button.
4. System redirects to 3PAS portal.
5. City Operator enters login credentials.
6. System receives verification confirmation regarding authentication of City Operator.
7. City Operator has access to SCEMAS dashboard.

Secondary scenario:

4.1.i System fails to redirect to 3PAS portal.
4.1.ii System re-launches account creation/login page.

5.1.i City Operator enter wrong authentication details.
5.1.ii System fails authentication, prompts user to try again.

6.1.i The system received verification was unsuccessful.
6.1.ii Session timed out, system prompts to try again and re-directs City Operator to homepage.

VP2: Public user #2

1b. Public User opens SCEMAS application.
3b. Public User clicks on Login Page, selects "Public User" button
5b. Public User enters login credentials.

Secondary Scenario:

4.1.i System fails to redirect to 3PAS portal.
4.1.ii System re-launches account creation/login page.

5.1.i Public User enter wrong authentication details.
5.1.ii System fails authentication, prompts user to try again.

6.1.i The system received verification was unsuccessful.
6.1.ii Session timed out, system prompts to try again and re-directs public user to homepage.

VP3: Third Party Developer #3

1c. Third Party Developer opens SCEMAS application.
3c. Third Party Developer clicks on Login Page, selects on "Third Party Developer" button.
5c. Third Party Developer enters login credentials.

Secondary Scenario:

4.1.i System fails to redirect to 3PAS portal
4.1.ii System re-launches account creation/login page.

5.1.i Third Party Developer enters wrong authentication details
5.1.ii System fails authentication, prompts user to try again.

6.1.i The system received verification was unsuccessful.
6.1.ii Session timed out, system prompts to try again and re-directs third party developer to homepage.

VP4 : System Administrator #4

1d System Administrator opens SCEMAS application.

3d. System Administrator clicks on Login Page, selects “System Administrator” button.
5d. System Administrator enters login credentials.

Secondary Scenario:

4.1.i System fails to redirect to 3PAS portal.
4.1.i System re-launches account creation/login page.

5.1.i System Administrator enter wrong authentication details.
5.1.ii System fails authentication, prompts user to try again.

6.1.i The system received verification was unsuccessful.
6.1.ii Session timed out, system prompts to try again and re-directs system administrator to homepage.

VP5 : Incident Response Team #5

1e. Incident Response Team opens SCEMAS application.
3e. Incident Response Team clicks on Login Page, selects on “Incident Response Team” button.
5e. Incident Response Team enters login credentials.

Secondary Scenario:

4.1.i System fails to re-direct to 3PAS portal.
4.1.ii System re-launches account creation/login page.

5.1.i Incident response team enters wrong authentication details.
5.1.ii System fails authentication, prompts user to try again.

6.1.i The system received verification was unsuccessful.
6.1.ii Session timed out, system prompts to try again and re-directs incident response team to homepage.

Global Scenario:

Pre-condition - Viewpoints have an existing account with SCEMAS
Main Success Story

1. User opens SCEMAS platform.
2. System launches Login/Account Creation Page.
3. The user navigates to the login page and selects the appropriate role.
4. System redirects to 3PAS portal.
5. User enters authentication details.
6. System receives verification confirmation regarding authentication of user.
7. Users have access to the SCEMAS dashboard.

Secondary scenario:

4.1.i System fails to re-direct to 3PAS portal.
4.1.ii System re-launches account creation/login page.

5.1.i User enters wrong authentication details.
5.1.ii System fails authentication, prompts user to try again.

6.1.i The system received verification was unsuccessful.
6.1.ii Session timed out, system prompts to try again and re-directs user to homepage.

BE6: Defines/update alert rules

Pre-Condition:
System Administrator has a verified account with SCEMAS
System Administrator has authorization to create or update alerts.
Incident Response Team is subscribed by default to any modifications made to alert threshold rules.

VP1: City Operator #1
N/A
VP2: Public User #2
N/A
VP3: System Administrator #3

1. System Administrator logins into dashboard and selects "Alerts" from dashboard.
2. The System displays active alerts currently in place. Displays options to update or create new alerts.
3. System Administrator clicks on "create new alert" button.
4. System displays alert configuration panel.
5. System Administrator defines alert rules such as alert type, sensor thresholds and region.
6. The system saves alerts, logs activity and redirects to alerts dashboard.
7. System admin activates alert on alert dashboard.
8. The system begins implementing alert rules and logs activity.

Secondary scenario:
5.1.i System Administrator enters invalid input.
5.1.ii System displays error message and prompts to correct input.

6.1.i System unable to save alert.
6.1. ii. Display error message prompts system administration to try again.
6.1.iii System logs activity.

7.1.i System unable to active newly created alert.
7.1.ii System displays error message and keeps alert inactive state.
7.1.iii System logs activity.

7.1.iv Requests System Administrator to rectify provided error or try again later.

VP4: Developer #4

N/A

VP5: Incident Response Team #5

7b. System notifies the Incident Response Team to modification or creation of active alerts.

Secondary scenario:

7.1.i. System fails to send notification to Incident Response Team.
7.1. ii. System Administrator receives failure notification.
7.1.iii. System logs failure and retries later.

Global Scenario:
Pre-Condition:

System Administrator has a verified account with SCEMAS.
System Administrator has authorization to create or update alerts.
Incident Response Team is subscribed by default to any modifications made to alert threshold rules.

Main Success Story

1. System Administrator logins into dashboard and selects "Alerts" from dashboard.
2. The System displays active alerts currently in place. Displays options to update or create new alerts is provided.
3. System Administrator clicks on "create new alert" button.
4. System displays alert configuration panel.
5. System Administrator defines alert rules such as alert type, sensor thresholds and region.
6. The system saves alerts, logs activity and redirects to alerts dashboard.
7. System admin activates alert on alert dashboard.
8. The system begins implementing alert rules and logs activity.

Secondary scenario:

5.1.i System Administrator enters invalid input.
5.1.ii System displays error message and prompts to correct input.

6.1.ii System unable to save alert.
6.1.iii Display error message prompts system administration to try again.
6.1.iv System logs activity.

7.1.i System unable to active newly created alert.
7.1.ii System displays error message and keeps alert inactive state.
7.1.iii System logs activity.

7.1.iv Requests System Administrator to rectify provided error or try again later.

5 Non-Functional Requirements

5.1 Look and Feel Requirements

5.1.1 Appearance Requirements LF- A1.

Requirement:

Across all operator-facing interfaces, SCEMAS shall present city data and alerts such that critical alerts are visually distinguishable within 1 second of being rendered on the screen, using visual emphasis techniques such as colour contrast or iconography. On authenticating into SCEMAS, the dashboard shall immediately display a modal window with immediate pending alerts to ensure user awareness of city ongoings.

Rationale:

Rapid visual identification of critical information is necessary for effective situational awareness in monitoring systems. Similar real-time monitoring platforms emphasize immediate visual differentiation of abnormal states to support operator response time [1].

5.1.2 Style Requirements LF-S1.

Requirement:

SCEMAS shall maintain a consistent visual style across all interfaces, including consistent use of colours, typography, and iconography without constraint-specific interface layouts or components.

Rationale:

A consistent visual style reduces user confusion and lowers the effort required to learn and navigate the system.

5.2 Usability and Humanity Requirements

5.2.1 Ease of Use Requirements UH-EOU1.

Requirement:

All users shall be able to access current conditions and view active alerts within no more than 3 interactions from initial login.

Rationale:

Limiting the number of required interactions reduces task completion time while minimizing the operation friction during both routine and emergency use cases.

5.2.2 Personalization and Internationalization Requirements UH-PI1.

Requirement:

SCEMAS shall allow authorized users to configure at least one personalized view (a view of alert information) such as filtering by alert type and/or severity. Multilingual / Internationalization requirements shall be supported for the initial release of SCEMAS.

Rationale:

Limited personalization increases usability while allowing development effort to remain focused on critical system functionality.

5.2.3 Learning Requirements UH-L1.

Requirement:

A new authorized operator shall be able to perform critical monitoring tasks, including viewing sensor data and acknowledging alerts, within 30 minutes of first interaction without formal training.

Rationale:

Efficient and ease-of-use priorities for users ensures that onboarding time in minimal. Allowing each operator to intuitively interact with the platform.

5.2.4 Understandability and Politeness Requirements UH-UP1.

Requirement:

All system-generated messages and alerts shall be presented using plain language with no message exceeding language used in secondary schools.

Rationale:

Using plain languages improves comprehension and mitigates the risk of misinterpretation, especially during high-stress situations

5.2.5 Accessibility Requirements UH-A1.

Requirement:

Operator-facing interfaces shall meet WCAG 2.1 level AA colour contrast requirements for text and critical visual elements.

Rationale:

Adherence to WCAG 2.1 Level AA guidelines improves accessibility for users with visual impairments and is widely adopted as a baseline standard for accessible software systems. [2]

Accessibility Requirements UH-A2.

Requirement:

SCEMAS shall be backwards compatible with legacy devices, included but not limited to all commonly used operating systems and device types.

Rationale:

Ensuring that the user-facing portal is available to all citizens is critical to the onboarding of SCEMAS and providing continued benefit to each user.

5.3 Performance Requirements

5.3.1 Speed and Latency Requirements PR-SL1.

Requirement:

The system shall process incoming sensor readings and make them available for alert evaluation within 900 ms of receipt under normal operating conditions.

Rationale:

Low end-to-end latency is required to support timely detection of abnormal environmental conditions in real-time monitoring systems.

5.3.2 Safety-Critical Requirements PR-SC1.

Requirement:

For safety-critical alerts classified as high severity, the system shall generate and display the alert within 2 seconds of threshold violation detection.

Rationale:

Delayed delivery of safety-critical alerts can lead to delayed response and increased risk in emergency situations.

5.3.3 Precision or Accuracy Requirements PR-PA1.

Requirement:

Aggregated data presented by the system shall preserve sensor measurement values with not more than +/- 1% deviation from the raw ingested data. This means rounding or the modification of underlying data types on SCEMAS shall not occur.

Rationale:

Accurate representation of city conditions is required to support reliable decision making and analysis.

5.3.4 Reliability and Availability Requirements PR-RA1.

Requirement:

The system shall maintain a minimum availability of 99.5% uptime, excluding scheduled maintenance periods.

Rationale:

High availability is required to ensure continuous monitoring and alerting capabilities in a smart city environment.

5.3.5 Robustness or Fault-Tolerance Requirements PR-RFT1.

Requirement:

The system shall tolerate the failure of individual sensor data sources without loss of functionality for remaining sensors and shall recover automatically within 60 seconds of failures.

Rationale:

Fault tolerance ensures continued operation despite partial system failures, which are common in distributed sensor networks.

5.3.6 Capacity Requirements PR-C1.

Requirement:

The system shall support ingestion and processing of data from at least 10,000 active sensors, each transmitting up to one reading per second.

Rationale:

Capacity requirements reflect anticipated deployment scale in a mid-sized smart city.

User Capacity Requirements PR-C2.

Requirement:

The system shall support 200,000 active users, being able to serve data and updates within 2 seconds. In the case of traffic spikes, the system can serve data within 10 seconds but maintain overall functionality.

Rationale:

The system should be able to handle growing city populations, including spikes during crisis events and updates. Typical metropolitan areas include extremely large populations, and thus SCEMAS should be able to accommodate a large number of users (200,00 was take as a estimated proportion of average resident populations). [3]

5.3.7 Scalability or Extensibility Requirements PR-SE1.

Requirement:

The system shall support horizontal scaling such that the number of supported sensors can be increased by at least 2 times without requiring architectural redesign.

Rationale:

Scalability allows the system to accommodate future growth in sensor deployment.

5.3.8 Longevity Requirements PR-L1.

Requirement:

The system shall be designed to operate continuously for a minimum expected lifetime of 15 years with routine software updates and maintenance.

Rationale:

Smart city infrastructure systems are expected to have long operational lifetimes to justify deployment cost.

5.4 Operational and Environmental Requirements

5.4.1 Expected Physical Environment OE-EPE1.

Requirement:

The system shall operate entirely in a server-based environment (deployed on the cloud) and shall not depend on any specific physical location or on-site hardware beyond external sensors.

Rationale:

A server-based deployment simplifies maintenance and supports centralized operation.

5.4.2 Requirements for Interfacing with Adjacent Systems OE-IA1.

Requirement:

The system shall interface with external sensor networks using standard communication protocols, i.e. MQTT.

Rationale:

Standardized interfaces improve interoperability with heterogeneous sensor vendors.

5.4.3 **Productization Requirements** OE-P1.

Requirement:

The system shall be deployable using documented configuration parameters without requiring code modification.

Rationale:

Productization requirements support repeatable deployment and operational consistency.

5.4.4 **Release Requirements** OE-R1.

Requirement:

Each system release shall include versioned documentation and release notes describing functional changes and resolved issues.

Rationale:

Clear release documentation supports operational transparency and maintenance.

5.5 Maintainability and Support Requirements

5.5.1 **Maintenance Requirements** MS-M1.

Requirement:

The system shall support routine maintenance activities, including configuration changes and software updates, without requiring full system downtime.

Rationale:

Reduced downtime improves availability and operational continuity.

5.5.2 **Supportability Requirements** MS-S1.

Requirement:

The system shall provide diagnostic logs for ingestion, alerting, and system errors, retained for a minimum of 30 days.

Rationale:

Diagnostic logging supports troubleshooting and incident investigation.

5.5.3 Adaptability Requirements MS-A1.

Requirement:

The system shall allow new sensor types to be integrated through configuration changes rather than core system redesign.

Rationale:

Adaptability supports evolving sensor technologies and future expansion.

5.6 Security Requirements

5.6.1 Access Requirements SR-AC1.

Requirement:

The system shall enforce role-based access control (RBAC), restricting system functions based on authenticated user roles.

Rationale:

Access control prevents unauthorized interaction with sensitive system functions.

5.6.2 Integrity Requirements SR-INT1.

Requirement:

All sensor data and system communications shall be protected against unauthorized modification using cryptographic integrity mechanisms.

Rationale:

Data integrity is required to ensure trustworthiness of sensor measurements.

5.6.3 Privacy Requirements SR-P1.

Requirement:

The system shall not store or process personally identifiable information (PII). 3PAS will handle this for feature-based needs.

Rationale:

Avoiding PII reduces privacy risk and regulatory burden.

5.6.4 **Audit Requirements** SR-AU1.

Requirement:

The system shall record audit logs for all authentication events and configuration changes, retained for at least 90 days.

Rationale:

Audit logging supports accountability and post-incident analysis.

5.6.5 **Immunity Requirements** SR-IM1.

Requirement:

The system shall reject malformed or unauthorized requests without affecting availability for legitimate users.

Rationale:

Immunity requirements reduce the impact of misuse or malicious input.

5.7 Cultural and Political Requirements

5.7.1 **Cultural Requirements**.

Requirement:

System must support multiple languages. CP-CP1

Rationale:

This enables users from diverse backgrounds to all have equitable access to SCEMAS. Ensure that critical information regarding safety and environmental conditions are delivered in an efficient manner and eliminate misunderstandings that can arise from language barriers. Furthermore, increase user engagement by offering inclusive system.

Requirement:

System must support users with visual and hearing impairments. CP-C2

Rationale:

Ensures that every person in the community has fair access to SCEMAS software. Users with hearing impairments, color blindness and other disabilities should still be able to receive important critical safety information and alerts. SCEMAS ensures to support vulnerable target users.

Requirement:

Software should not include any offensive signs or imagery. CP-C3

Rationale:

SCEMAS dashboard should not display or utilize language, imagery or signs that can be considered insensitive and offensive. The system must account for the diversity amongst its user population and ensure to remain respectful to all communities.

Requirement:

The system should maintain system transparency by informing the public when sensors are under maintenance or unavailable. CP-C4

Rationale:

This notifies users if data coverage is low or unavailable in certain regions/zones. This clear communication reduces misinformation, misinterpretation and incorrect conclusions of the information displayed. Furthermore, it builds credibility and increases users' trust.

5.7.2 Political Requirements

Requirement:

The software must allow the municipal government to publish public announcements, issue official notices and safety briefings on the dashboard when required. CP-P1

Rationale:

This would overall increase user's trust and reliance on the system as the SCEMAS dashboard becomes one of the communication pathways the government can reach to its citizens.

Requirement:

The system should allow authorized government authorities to use data collected by SCMEAS for discussion, planning key initiatives and policy development. CP-P2

Rationale:

This expands SCEMAS's user base as it now supports municipal government. The system will provide collected data from sensors to help justify decisions and create policies. Hence this would not only increase reliance for government bodies but also user trust from the public.

5.8 Legal Requirements

5.8.1 Compliance Requirements

Requirement:

No personal information (PII) should be collected, processed, or stored by the system, and all data collected by third parties must be protected. LR-COMP1.

Rationale:

This is PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 1 – Accountability [4]

Requirement:

The purpose for all personal information (PII) being collected (by a 3rd party – not SCEMAS) must be identified before or at the time it is being collected. LR-COMP2

Rationale:

This is PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 2 – Identifying Purposes [4]

Requirement:

The consent and knowledge of any user (despite their clearance) for all personal information (PII) being collected (including information collected by a 3rd party) is required. LR-COMP3

Rationale:

This is PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 3 – Consent [4]

Requirement:

Only the personal information (PII) that is needed to accomplish a legitimate purpose informed to the user must be collected by a 3rd party. Nothing Aside from that. LR-COMP4

Rationale:

This is PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 4 – limiting collection [4]

Requirement:

personal information (PII) collected (including information collected by 3rd parties) must be used or disclosed only for the purpose it was identified for, except if the user consents or under any legal obligations. LR-COMP5.

Rationale:

This is PIPEDA (Personal Information Protection and Electronic Documents Act) Fair Information Principle 5 – Limiting Use, Disclosure, and Retention [4]

5.8.2 Standards Requirements

Requirement:

Apps must not log sensitive personal information (PII) to the system log. Any dynamic data included in log messages must be marked with the appropriate privacy level. LR-STD1

Rationale:

This is standard for companies like Apple to help protect user privacy, the unified logging system provides a mechanism for redacting sensitive information. [9]

Requirement:

Push Notifications must not be required for the app to function, and should not be used for advertising, promotions, or direct marketing purposes unless customers have explicitly opted in to receive them via consent language displayed in the app's UI. LR-STD2

Rationale:

This is a standard for app quality for companies like Apple. [10]

Requirements:

Any personal or sensitive data should not be logged to system log or any sort of log for the application. LR-STD3

Rationale:

Rationale: This is a standard for app quality for companies like Apple. [9]

6 Innovative Feature

The chosen innovative feature to include in SCEMAS is the addition of an Incident Response Team. The purpose of this team is to allow subscribed users to receive immediate feedback and/or help in the case of an alert, (e.g, natural disaster or forecasted trends). Information from the Incident Response Team is sent alongside typical SCEMAS alerts but differs in that it gives context-based aid to better help each individual. As well as that, the Incidence Response Team may send season-based advice to all users of what they may want to watch out for and how they can be prepared in the case of an alert.

This feature benefits the product as unlike the main structure of the app that is constructed to identify anomalies in the environment, this covers the aspect of what individuals can do to protect themselves from identified anomalies. This feature improves loss mitigation for the city, leading to a reduction of lives lost, damages, and potential lawsuits. Furthermore, this feature promotes smart city initiatives as it's responsive to citizen's needs. Overall, the addition of this feature will lead to an increase in users as they can rely on this feature in times of panic.

Innovative features discussed are as follows but not limited to:

- Live reporting (Done by users)
- ML Forecasting
- Report Bounties (For Users)
- Support Hot Line (for non-emergency questions)
- Weekly informative podcast
- Natural Disaster Sim
- Computer vision feature to identify what to take with you during alerts

Division of Labour

Colin:

- 1. Introduction
- 5 – 5.6 inclusive
- Wrote Section 6
- (Brainstorm Innovative feature)
- Full document revision with team

Sachika

- Wrote Business Events 4, 5 and 6

- Created Use case Diagram
- Wrote section 5.7
- Brainstormed innovative feature
- Full document revision with team

Dharav

- Business Events 1,2,3,
- 5.8
- Brainstormed innovative features
- Reviewed and made small changes with Colin to 5 – 5.6
- Full document revision with team

Ali Zain

- Wrote Section 2 (2.1–2.6).
- Revised Section 1.2 (Scope).
- Updated Section 1.3 (Definitions)
- Expanded Section 1.4 (References).
- Contributed to Section 6 (innovative feature).
- Fixed formatting and consistency issues across the document and completed a full document revision with the team.